

■ Speed, Distance & Time

Grade 3 Maths — Lesson Guide

■ What Are Speed, Distance & Time?

■ Speed

How fast something moves

e.g. Running fast = high speed

■ Distance

How far you travel

e.g. Home to school = 2 km

■ Time

How long the journey takes

e.g. Walk to school = 10 min

Time →

■ The Magic Formula

$$\text{Speed} = \text{Distance} \div \text{Time} \quad (S = D \div T)$$

Find Speed

Find Distance

Find Time

$$S = D \div T$$

$$D = S \times T$$

$$T = D \div S$$

■ Worked Examples

Example 1: Walking to School ■

Given: Riya walks 2 km in 30 minutes (= 0.5 hr)

Find: What is her speed?

Working: Speed = Distance $\div$ Time = $2 \div 0.5$

■ Speed = 4 km/hr

Example 2: Cycling to the Park ■

Given: Arjun cycles 6 km in 30 minutes (= 0.5 hr)

Find: What is his speed?

Working: Speed = Distance $\div$ Time = $6 \div 0.5$

■ Speed = 12 km/hr

Example 3: Running Race ■

Given: Priya runs 100 m in 20 seconds

Find: What is her speed?

Working: Speed = Distance $\div$ Time = $100 \div 20$

■ Speed = 5 m/sec

■ Real-Life Problems

■ School Bus

Speed = 40 km/hr, Distance = 20 km Time = $20 \div 40 = 0.5$ hr = 30 min

■ Delhi to Jaipur

Distance = 280 km, Time = 4 hours Speed = $280 \div 4 = 70$ km/hr

■ Quick Quiz

Q1: A car travels 60 km in 2 hours. What is its speed?

A) 20 km/hr

B) 30 km/hr

C) 40 km/hr

■ Answer: B) 30 km/hr

Q2: You run 200 m in 40 seconds. What is your speed?

A) 3 m/sec

B) 5 m/sec

C) 8 m/sec

■ Answer: B) 5 m/sec

Q3: A bus takes 3 hours to travel 150 km. What is its speed?

A) 40 km/hr

B) 50 km/hr

C) 60 km/hr

■ Answer: B) 50 km/hr

■ What Did We Learn?

■ Speed tells us how fast we go

■ Distance tells us how far we travel

■ Time tells us how long it takes

■ Magic Formula: $\text{Speed} = \text{Distance} \div \text{Time}$

■ Bonus Speed Challenge — Try at Home!

1. Mark 10 metres on the ground
2. Run from start to finish
3. Ask a friend to time you with a stopwatch
4. Calculate: $\text{Speed} = 10 \text{ m} \div \text{your time (in seconds)}$
5. Share your speed with your classmates!

Great Job! You can now solve Speed, Distance & Time problems! ■